

SALTY SEAS

Water is very good at dissolving substances, such as the minerals that form rocks. Water flowing off the land in rivers contains many of these dissolved minerals, known as salts, as well as particles of sand and mud. They are carried down to the sea, where they have built up over billions of years. Most of the salt is sodium chloride, which is the same as table salt. This is why seawater tastes salty.

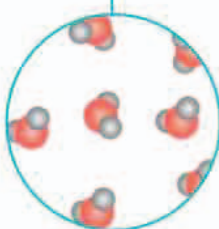


▲ SALTY LAKES

In hot regions, water flowing off the land can evaporate to leave salt crystals such as these on the edge of a salty lake. In oceans, the salt is dissolved and invisible.

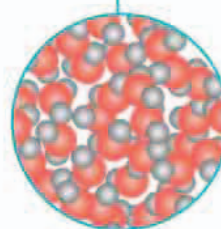
SOLID, LIQUID, AND GAS

At very low temperatures, water molecules lock together to form solid ice. At higher temperatures, the molecules come unstuck, so they drift apart to form water vapor. The temperature difference between freezing and evaporating is so small that all three states—ice, water, and water vapor—can exist together in the same place, at the same time. This is a unique property of water.



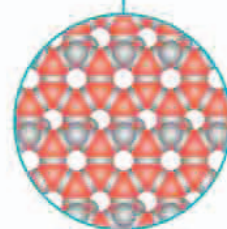
▲ GAS

When water turns to water vapor (a gas), the molecules drift apart and float in the air.



▲ LIQUID

In liquid water, the molecules cling together, but are able to move. This allows water to flow.



▲ SOLID

When water freezes, the molecules move apart and lock into a solid 3-D pattern, forming ice.

CHEMISTRY OF LIFE

As well as the minerals that make it salty, ocean water also contains other dissolved substances, including carbon, oxygen, nitrogen, phosphorus, calcium, and iron. These are essential ingredients of complex molecules such as proteins, which are vital to life in all its forms. This makes ocean water an ideal habitat—fossil evidence suggests that the first life on Earth developed in the oceans. The seas and oceans are still teeming with an incredible array of animals.

